

Rewritten Chain of Thought for the Solution to the Two-Minus-Sector Water-Wave Amplitude A_n

Abstract

This document contains a rewritten summary of the chain of thought for the closed-form solution of the tree-level n -point on-shell amplitude A_n for one-dimensional deep-water surface gravity waves in the two-minus sector $\sigma = (-1, -1, +1, \dots, +1)$, in PDF form.

Rewritten Chain of Thought

The object is the tree amplitude A_n produced by the Berends–Giele recursion in `OnShellBG.m`, evaluated on the resonant manifold

$$\sum_{i=1}^n \omega_i = 0, \quad \sum_{i=1}^n \sigma_i \omega_i^2 = 0,$$

in the two-minus sector $\sigma = (-1, -1, +1, \dots, +1)$. The dispersion law $\omega_i^2 = g |k_i|$ fixes the momenta as $k_i = \sigma_i \omega_i^2 / g$. I set $g = 1$ until the very end, where dimensional analysis restores it. Legs 1, 2 are the two negative- σ legs and legs $3, \dots, n$ the positive ones. The hint promises a piecewise homogeneous polynomial in the frequencies, and the task is to produce it.

Before generating any data it pays to ask where non-polynomiality could possibly enter. The recursion is assembled from rational kernels, a propagator $-i/(\omega_S^2/|k_S| - g)$, and the magnitude $\text{mag}[k] = |k|$. The only non-rational operation anywhere is that absolute value, and it is applied exclusively to *momenta*, never to frequencies. An external momentum is $k_i = \sigma_i \omega_i^2$, so $|k_i| = \omega_i^2$ is unambiguous. The recursion, however, forms *internal* momenta

$$k_S = \sum_{i \in S} \sigma_i \omega_i^2$$

on subsets S , and $|k_S| = |\sum_{i \in S} \sigma_i \omega_i^2|$ is genuinely sign-dependent. This is the only place a piecewise structure can come from: the chambers must be the regions on which every k_S keeps a fixed sign. I keep this in reserve and start measuring.

A handful of exact rational evaluations settle the gross features. Every A_n comes out purely imaginary, and rescaling all frequencies by λ multiplies the value by $\lambda^{2(n-2)}$ — at $n = 5$ the ratio is $64 = 2^6$, at $n = 6$ it is $256 = 2^8$. So

$$A_n = -i P_n, \quad P_n \text{ real, homogeneous of degree } 2(n-2),$$

and that degree is the first hard constraint on any guess.

The $n = 4$ evaluation refuses to return a number; it is $0 \cdot \infty$. The kinematic solver, which fixes two frequencies from the conservation laws, has no freedom at $n = 4$ but to land on $\omega_4 = -\omega_2$ (and $\omega_1 = -\omega_3$), forcing the internal momentum $k_{\{2,4\}} = -\omega_2^2 + \omega_4^2 = 0$. A vanishing internal momentum sends the propagator to zero while a kernel develops a $0/0$, and the symbolic engine

cannot resolve the product. Choosing a different pair of solved legs only moves the zero to another channel. The conclusion is structural rather than accidental: the entire $n = 4$ two-minus variety lies on a wall where some $k_S = 0$. I set $n = 4$ aside as a boundary case to be recovered later, and work at $n \geq 5$, where the interior is non-empty.

Permuting the two minus legs, or permuting the plus legs among themselves, preserves both conservation laws and so maps on-shell points to on-shell points. Direct evaluation shows A_n is invariant under all of these: a generic point and all $2 \cdot (n - 2)!$ of its images return one and the same value. Thus A_n is symmetric under $S_2 \times S_{n-2}$.

This symmetry seems to hand me the answer. On the constraint surface the two minus-leg symmetric functions reduce to those of the plus legs: writing P_k for the k -th elementary symmetric polynomial of $\omega_3, \dots, \omega_n$, conservation gives $\omega_1 + \omega_2 = -P_1$ and $\omega_1 \omega_2 = P_2$. Hence any $S_2 \times S_{n-2}$ -symmetric polynomial of degree $2(n - 2)$, restricted to the surface, is a weighted-homogeneous polynomial in $P_1, \dots, P_{n-2}$ alone — for $n = 5$ that is exactly seven monomials. I generate points, set up the 7×7 linear system, and solve.

It does not fit. Not approximately: with exact rational points and a full-rank system the residual on held-out points is of order 10^{11} . Suspecting a degenerate sample, I widen and diversify the points; the residual stays enormous. This is a genuine contradiction. The function is symmetric (checked directly), homogeneous of a definite degree (checked directly), and the hint says polynomial — yet it is provably not the restriction of any symmetric polynomial, since such a restriction would have to lie in the seven-dimensional span that the data refute.

There is only one way out, and it is the chamber idea I had parked. A symmetric *function* need not be a single symmetric polynomial if it is merely *piecewise* polynomial: the global symmetry can be realised by gluing together pieces that are individually *not* symmetric, with the symmetry group permuting the pieces. The walls are the sign changes of the internal momenta k_S . Within one chamber every absolute value resolves to a fixed sign, so the amplitude is a single rational function and — by the hint — a polynomial; but across a wall $|k_S|$ flips to $-k_S$ and the polynomial can change. The piece in any one chamber carries no obligation to be symmetric, only to glue continuously to its neighbours. The fit failed because it demanded a global symmetry that no single piece possesses.

To see one piece cleanly I collapse to a line: fix ω_3, ω_4 at rational values, let $\omega_2 = t$ vary, and let the solver give ω_1, ω_5 as functions of t . A symbolic attempt drowns — the nested absolute values of rational functions generate expressions large enough to exhaust memory — so instead I evaluate the exact rational $A_5(t)$ at many rational t inside a single chamber and reconstruct the function. Multiplying by the apparent denominator $\omega_2 + \omega_3 + \omega_4 = 13 + t$ clears it to a polynomial,

$$A_5(t) (13 + t) = 208 t^6 + 2128 t^5 = 16 t^5 (13t + 133),$$

and since the solver returns $\omega_1 = -(13t + 133)/(13 + t)$, the denominator cancels against the numerator factor, leaving the clean monomial

$$A_5 = -16 \omega_1 \omega_2^5$$

on this chamber. The factor of ω_1 was hidden inside the apparent pole; the result depends on the plus legs not at all. Checking against scattered points of the same chamber — wildly different plus-leg data, the identical monomial — confirms it. And the chamber is recognisable: it is the region in which ω_2 is the unique smallest frequency, equivalently the one in which $k_{\{2\}}$ is the *only* negative subset momentum.

The monomial $-16 \omega_1 \omega_2^5$ is wildly asymmetric in the two minus legs, exactly as it should be: the chamber “ ω_2 smallest” is not invariant under $\omega_1 \leftrightarrow \omega_2$ — that swap leaves the chamber — so the

piece living on it is free to be asymmetric. The same monomial recurs at every n in the smallest chamber, with a clean power of two for a coefficient; evaluating at $n = 6, 7$ gives

$$A_n = -2^{n-1} \omega_1 \omega_2^{2n-5} \quad (\omega_2 \text{ smallest}),$$

whose exponents 1 and $2n - 5$ sum to the required $2(n - 2)$.

Now I let ω_2 grow and watch the amplitude cross walls. As ω_2^2 rises past each partial sum $q_S = \sum_{i \in S} \omega_i^2$ of plus-leg squares, the subset momentum $k_{\{2\} \cup S} = -\omega_2^2 + q_S$ changes sign. Reconstructing the polynomial on each successive chamber and differencing neighbours, the jump across the wall $\omega_2^2 = q_S$ is always

$$\Delta_S = (-1)^{|S|+1} 2^{n-1} \omega_1 \omega_2 (\omega_2^2 - q_S)^{n-3},$$

which vanishes on the wall (the function is continuous) and switches on only beyond it. Starting from the smallest chamber — where only $S = \emptyset$ is active and the piece is $-2^{n-1} \omega_1 \omega_2 (\omega_2^2)^{n-3} = -2^{n-1} \omega_1 \omega_2^{2n-5}$ — and adding the jump for every wall already crossed, the amplitude assembles into a single closed form,

$$A_n = -i 2^{n-1} \omega_1 \omega_2 \sum_{\substack{S \subseteq \{3, \dots, n\} \\ \omega_2^2 > q_S}} (-1)^{|S|+1} (\omega_2^2 - q_S)^{n-3},$$

the sum running over precisely those subsets whose wall has been crossed. Equivalently, with the truncated power $(x)_+ = \max(x, 0)$, it is the spline

$$A_n = -i 2^{n-1} \omega_1 \omega_2 \sum_{S \subseteq \{3, \dots, n\}} (-1)^{|S|+1} [(\omega_2^2 - q_S)_+]^{n-3}.$$

A reassuring instance: in the chamber $\omega_3^2 + \omega_4^2 < \omega_2^2$ at $n = 5$, where all of $\emptyset, \{3\}, \{4\}, \{3, 4\}$ are active, the four truncated powers combine, the ω_4^4 and ω_2^2 terms cancel, and only $-32 \omega_1 \omega_2 \omega_3^2 \omega_4^2$ survives — again a clean monomial, now in the four small legs.

Two facts make the formula consistent rather than merely fitted. First, the exponent is $n - 3$ while the number of plus legs is $n - 2$, and the $(n - 2)$ -fold alternating sum of a polynomial of degree $n - 3$ vanishes identically,

$$\sum_{S \subseteq \{3, \dots, n\}} (-1)^{|S|} (x - q_S)^{n-3} = 0 \quad \text{for all } x,$$

which I verified symbolically for $n = 4, \dots, 7$. This is why the spline closes up: the jumps across a full nested set of walls sum to zero, leaving nothing at the far end. Second, the same identity disposes of an apparent ambiguity. I wrote the threshold with ω_2^2 , but S_2 -symmetry demands it not matter whether I threshold on ω_2^2 or ω_1^2 . On shell $\omega_1^2 + \omega_2^2 = q_{\{3, \dots, n\}}$ and $q_{S^c} = q_{\{3, \dots, n\}} - q_S$; fed through the vanishing identity these relate the truncated sum built on ω_1^2 to the one built on ω_2^2 , and numerically the two agree at every point and in every chamber, including those where ω_1 is the smaller leg. The formula is genuinely well defined on the symmetric variety, not an artefact of labelling.

The boundary case now falls out. At $n = 4$ the variety is forced onto the wall $\omega_2^2 = \omega_4^2$, so the Berends–Giele evaluation sits exactly on a discontinuity of the absolute value and returns $0 \cdot \infty$. But the spline is continuous there, with the definite value $-i 8 \omega_1 \omega_2^3$, and approaching the wall through nearby off-resonant points the recursion converges to precisely this number. The $0 \cdot \infty$ is a removable singularity whose value the closed form supplies.

Finally the coupling. Restoring g , the amplitude is not invariant: comparing $g = 1$ with $g = 2$ shows $A_n \propto g^{-(n-3)}$. This is forced once one sees that the natural variable inside the bracket is the magnitude $|k_j| = \omega_j^2/g$ rather than ω_j^2 ; written in terms of $|k_j|$ the factors of g cancel, and only the explicit $g^{-(n-3)}$ from re-homogenising remains,

$$A_n = -i \frac{2^{n-1}}{g^{n-3}} \omega_1 \omega_2 \sum_{S \subseteq \{3, \dots, n\}} (-1)^{|S|+1} [(\omega_2^2 - q_S)_+]^{n-3}, \quad q_S = \sum_{i \in S} \omega_i^2.$$

What remains is to be sure the guess is not merely a good fit on the line where it was found. Calling the recursion on one representative of every chamber reachable by sweeping ω_2 — at $n = 7$ these range from the one-term smallest chamber up to chambers whose polynomial carries sixteen truncated-power terms — the closed form reproduces the exact rational Berends–Giele value with identically zero residual; the same holds at hundreds of random points and at explicit configurations with ω_1 smallest, for $n = 5, 6, 7$, together with the limiting agreement at $n = 4$. Zero residual in exact arithmetic is stronger than any tolerance: the piecewise polynomial *is* the amplitude.

In hindsight the structure is natural. The combination $\sum_S (-1)^{|S|} [(x - q_S)_+]^m$ is the standard inclusion–exclusion / truncated-power building block of a one-dimensional spline with knots $\{q_S\}$; here the evaluation point is $x = \omega_2^2$, the knots are the partial sums of plus-leg squares, and the two minus legs supply the prefactor $\omega_1 \omega_2$. The amplitude is, up to that prefactor and the power of two, a spline in the single variable ω_2^2 against the knot set $\{q_S\}$; its polynomial pieces are the kinematic chambers, and the vanishing of the full alternating sum — degree one below the number of knots — is exactly the condition that the pieces glue into one globally symmetric, globally continuous function.